Alexandre Autran

Mathematics and Machine Learning Student

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Profile

I am a third-year mathematics and machine learning student at the École Normale Supérieure (ENS) de Rennes. My research focuses on stochastic processes and optimization, with applications to machine learning, time-series modeling, and quantitative analysis. I combine theoretical and computational approaches, with particular interests in graph learning and generative models.

Education

2024-2027 **Student ("Élève normalien") in the Mathematics Department**
Admitted to the ENS through a competitive entrance examination.
Four-year ENS diploma pursued over three years (accelerated track).

2026-2027 **Research Year in Machine Learning**, École Normale Supérieure Paris-Saclay, France
Final year of the ENS Diploma (research track, "Prelab" ENS).

2025-2027 **MSc in Fundamental Mathematics**, École Normale Supérieure de Rennes, France
M2: Highest honors (17/20 in the S1, completed alongside M1).
M1: Highest honors (17.6/20), rank: 5/100+.

2024-2025 **BSc in Mathematics, minor in Computer Science**, École Normale Supérieure de Rennes, France
L3: High honors (15.4/20).

2022-2024 **BSc in Mathematics, minor in Computer Science**, University of Rennes, France
L2: Highest honors (18/20), rank: 3/100+.
L1: Highest honors (18.5/20), rank: 1/100+.

Publications [Full list]

[1] **A. Autran**, Singular Tracking of Fractional Brownian Motion under Proportional Costs, arXiv preprint, 2026. Under review at *Stochastic Processes and their Applications*.

[2] **A. Autran**, Conditional Fluctuations of Mean-Field Systems with Stable Common Noise, arXiv preprint, 2026. Under review at *Electronic Journal of Probability*.

[3] **A. Autran** et al., Multi-Dimensional Charge Diffusion-Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles, *Applied Sciences*, 16(11), 5551, 2026.

Research Experiences [Full list]

Sep-Feb 2027 **Research Internship**
High-Dimensional Density Clustering Using Local Unimodality.
Supervised by Argyris Kalogeratos at Centre Borelli, ENS Paris-Saclay, France.

Jun-Aug 2026 **Research Internship**
Optical Modeling of Thermochromic VO_2-Based Oxide Nanocomposites.
Supervised by Nguyen Ngoc Duy and Amaury Baret at University of Liège, Belgium.

Apr-Jun 2026 **Research Internship**
Numerical Simulation and Optimal Control of Spatial Lotka-Volterra Systems.
Supervised by Martin Gander and Bastien Chaudet-Dumas at the University of Geneva, Switzerland.

Jun-Aug 2025 **Research Internship**
Lattice-Based Post-Quantum Cryptography.
Supervised by Pierre-Alain Fouque and Aurore Guillevic at Inria, France.

Apr-Jun 2025 **Research Internship**
Algebraic Counting of the Real Roots of Polynomials.
Supervised by Marie-Françoise Roy, CNRS, France.

GitHub Projects [Full list]

2026 **Comparing Dense and Sparse Neural Models for Constrained Graph Generation**
PyTorch implementation of diffusion-based graph generation with hard connectivity and edge-budget guarantees, with a focus on structural validity and efficient generation.

2026 **Time-Series Statistical Arbitrage under Transaction Costs and Forecast Uncertainty**
Built and tested a statistical arbitrage strategy on six cryptocurrency perpetual futures, combining return forecasting, cost-aware portfolio optimization, and walk-forward validation with bootstrap inference.

Other Activities

2024-2025 **ENS Oral Examinations Preparation**
Supervision of second-year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures (Ulm, Paris-Saclay, Rennes, and Lyon).

2022-2025 **Teaching in Mathematics**
Tutoring for final-year high school students at Lycée Chateaubriand in Rennes, and for first- and second-year BSc students at the University of Rennes.

2023-2024 **BSc Student Representative**
Representative on the improvement council of the Mathematics Department of the University of Rennes.

Before 2022 **High School Competitions**
Participation in the Concours Général in Mathematics, Physics, and Philosophy, and the National High School Mathematics Olympiad, at Lycée Thiers in Marseille.

Technical Skills

Machine Learning: PyTorch, scikit-learn, ridge regression, gradient boosting, diffusion models.
Optimization: CVXPY, convex optimization, robust portfolio allocation, optimal control.
Scientific Computing: NumPy, pandas, MATLAB, R.
Programming & Tools: Python, C/C++, Java, SQL, Git, LaTeX.

Languages

French: native. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: fantasy and science fiction literature, chess.

Academic References

François Bolley – francois.bolley@ens-rennes.fr
Full Professor, IRMAR, ENS de Rennes.
Director of the Mathematics Department at ENS de Rennes.

Martin Gander – martin.gander@unige.ch
Full Professor, Section of Mathematics, University of Geneva.

Argyris Kalogeratos – argyris.kalogeratos@ens-paris-saclay.fr
Senior Research Scientist, Centre Borelli, ENS Paris-Saclay.